

Machine learning:

(Monk Cerb) d. 2
(Slide 13)

1. Bernoulli case $X \sim B(p)$

$$X: \Omega \rightarrow \mathbb{E} = \{1, 0\}$$

$$\text{law} \begin{cases} P(X=1) = p \\ P(X=0) = 1-p \end{cases}, 0 < p < 1$$

Algo.

1. $u = \text{RAND}$

$$2. \hat{X}(u) = \mathbb{1}\{u(u) \leq p\} = \mathbb{1}_{[0, p]}(u(u))$$

$$\hat{X} \sim X$$

Proof:

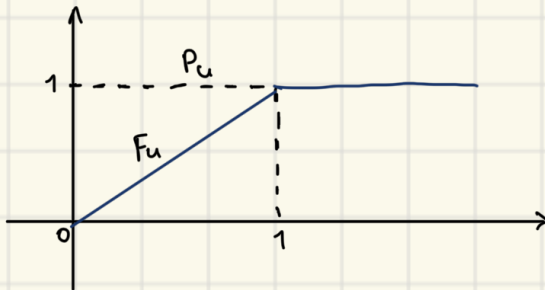
$$P(\hat{X}=1) = P(\mathbb{1}_{[0, p]} = 1) = \underline{P(u \leq p)} \rightarrow \text{random value}$$

Distribution function of u : $= F(u) = p$ as $0 < p < 1$

$$F_u(x) = P(u \leq x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x \geq 1 \end{cases}$$

* Important:

Density of random uniform variable



Any distribution function is between 0 and 1

2. Geometric & Poisson (Slide 13)

3. The inverse function method (Slide 15-16)

$$X: \Omega \rightarrow \mathbb{R}$$

General alg. for r.v. for distribution function F .

$$F(x) = P(X \leq x)$$

$$\left[\frac{P(X \leq b) = F(b) - F(a)}{(a, b)} \right] \rightarrow u \text{ can find the } p \text{ of a finite interval}$$

Example 1: $X \sim e(1)$ (exponential law), $\lambda=0, x \geq 0$

$$F(x) = 1 - e^{-\lambda x}, x \geq 0$$

$$y = F(x) = 1 - e^{-\lambda x}, x > 0, 0 < y < 1$$

$$e^{-\lambda x} = 1 - y$$

$$-\lambda x = \ln(1 - y)$$

$$F^{-1}(y) = x = -\frac{1}{\lambda} \ln(1 - y)$$

Example 2: Normal function

$$Z \sim N(0, 1)$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \text{ pdf}$$

$$\Phi = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du$$

, we do not have a closed form
 $\Rightarrow$ we can not use it in this case

The theory: When we replace a deterministic function with a random variable X , $\Rightarrow$ we have a new random variable, it is a uniform one.

$$F(X) \sim \text{Unif}(0, 1)$$

$$F^{-1}(u) \sim X$$

$\rightarrow$ if we reverse $F(y)$ and replace y with u and we get the initial variable X .

$$X = -\frac{1}{\lambda} \ln(1 - u) \rightarrow \text{General formula of exponential form of random variable}$$

Ex.: As $1 - u \sim w$

$$X(w) = -\frac{1}{\lambda} \ln(1 - w)$$

$$\lambda = 10^{-2} h^{-1}$$

First step of alg.: 1. $u(w) = 0.35$
 2. $X(w) = -\frac{1}{10^2} \ln(1 - 0.35) = 104 \rightarrow h$

ex. $X \sim W(\alpha, \beta)$

$$F(x) = 1 - e^{-\left(\frac{x}{\alpha}\right)^\beta}, x \geq 0$$

$$y = 1 - e^{-\left(\frac{x}{\alpha}\right)^\beta}$$

$$e^{-\left(\frac{x}{\alpha}\right)^\beta} = y - 1$$

$$\ln e^{-\left(\frac{x}{\alpha}\right)^\beta} = \ln(y - 1)$$

$$\left(-\frac{x}{\alpha}\right)^\beta = \ln(y - 1)$$

$$F^{-1}(x) = x = -\alpha \sqrt[\beta]{\ln(y - 1)} \rightarrow u$$

* **Exam:** To prove $1-u \sim u$ (fllw the same distribution function as u)

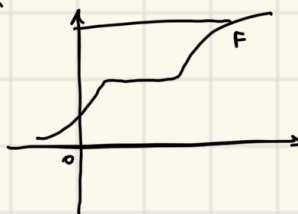
$$X(u) = L[-\ln(u(u))]^{1/\beta}$$

$$\Leftrightarrow \begin{matrix} X \sim Y \\ F_X \sim F_Y \end{matrix}$$

Inverse function method proof:

$$y = F_X^{-1}(u) \sim X$$

Suppose F_X is invertible



To be invertible it should be strictly increasing.

$$\begin{aligned} F_Y(x) &= P(Y \leq x) = P(F_X^{-1}(u) \leq x) \\ &= P(F_X(F_X^{-1}(u)) \leq F(x)) \quad \text{Id } u = u \\ &= P(u \leq F(x)) = F_u(F(x)) = F_X(x) \quad \forall x \end{aligned}$$

Rejection Method (Skip)

4. **Gaussian r.v. : The BOX-MULLER Method**

i.i.d (independent identity distribution)

$$\begin{aligned} &U_1 \text{ and } U_2 \text{ (both are produced randomly)} \\ \Rightarrow \begin{cases} U_1 = \text{RAND} \\ U_2 = \text{RAND} \\ (1-u \text{ by } u) \end{cases} &\text{also} \quad \begin{aligned} X &= \sqrt{-2 \ln U_1} \cos(2\pi U_2) \\ Y &= \sqrt{-2 \ln U_1} \sin(2\pi U_2) \end{aligned} \quad \text{are i.i.d.} \sim N(0,1) \end{aligned}$$

$$Z \sim N(0,1)$$

How to realize the r.v. $X \sim N(m, \sigma^2)$, $m = E(X)$, $\sigma^2 = \text{Var}(X)$

to make mean 0, $\text{Var}(ax+b) = a^2 \text{Var}(x)$

$$\frac{X-m}{\sigma} \sim Z \sim N(0,1) \Leftrightarrow X = \frac{m + \sigma Z}{1}$$

Step 1: $U_1 = \text{RAND}; U_2 = \text{RAND}$

Step 2: $Z = \sqrt{-2 \ln U_1} \cos(2\pi U_2)$ → The algorithm

Step 3: $X = \sigma Z + m$

5. Precision (Evaluation of Monte Carlo)

$$R_n = \frac{h(x_1) + h(x_2) + \dots + h(x_n)}{n} - E[h(x)]$$

→ Empirical mean → Theory mean

We define the precision by parameters ε and θ , which are small:

$$P(|R_n| \leq \varepsilon) \geq 1 - \theta$$

or

$$P(|R_n| \geq \varepsilon) \leq \theta$$

$$A, A^c \quad A \cup A^c = \Omega$$

$$P(A) = 1 - P(A^c)$$

(ε, θ) precision

* Define n in order to achieve this precision

Chebyshev inequality (slide 23)

General form:

Let $X: \Omega \rightarrow \mathbb{R}$, $E(X^2) < \infty \Leftrightarrow \text{Var}(X) < \infty$

$$P(|X - E(X)| > \varepsilon) \leq \frac{\text{Var}(X)}{\varepsilon^2}, \forall \varepsilon > 0$$

$$P(|R_n| \geq \varepsilon) \leq \frac{\text{Var}(h(X_1))}{n\varepsilon^2}$$

In order to achieve the precision (ε, θ) we must have:

$$\theta \geq \frac{\text{Var}(h(X_1))}{n\varepsilon^2} \Rightarrow \text{min value } n \geq \frac{\text{Var}(h(X_1))}{\theta\varepsilon^2}$$

$\text{Var}(h(X_1))$ can be estimated or bounded

To estimate $\text{Var}(h(X))$:

($m \sim 100$)

$$X_1, \dots, X_m$$

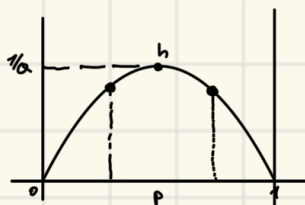
$$\frac{h(X_1), \dots, h(X_m)}{m} = \bar{h}_m$$

$$C = \text{Var}(h(X)) = \frac{1}{m} (h(X_i) - \bar{h}_m)^2$$

Example: $X_1, X_2, \dots, X_n$ be r.v i.i.d. $\sim B(p)$

Estimated the expectation by Monte Carlo, $\varepsilon = \theta = 0.01$, what's min. value of n

$$n \geq \frac{C}{\theta\varepsilon^2} = 250000$$

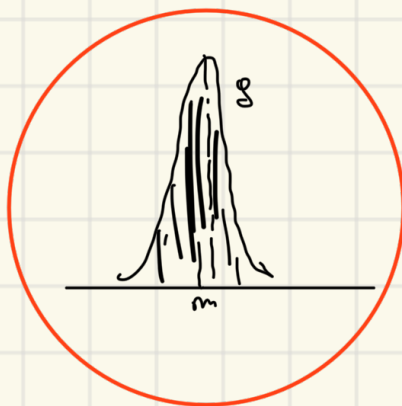
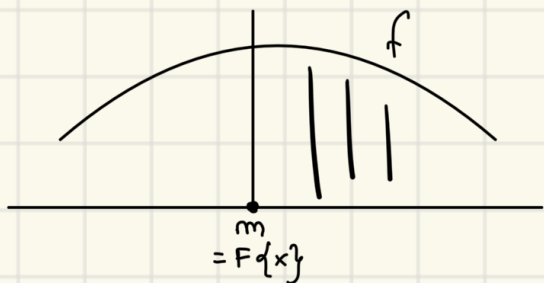


$$h(p) = p(1-p) \text{Var}$$

$$\frac{X_1 + \dots + X_n}{n} = \bar{X}$$

$$P(|\bar{X} - p| > 10^{-2}) \leq 10^{-2}$$

6. Acceleration or variance reduction



$$y = h(x), h \geq 0 \quad E[y] = E[h(x)] = \int_{\mathbb{R}} h(x) f(x) dx = \int_{\mathbb{R}} \left[h(x) \frac{f(x)}{g(x)} \right] g(x) dx = E_g \left[\frac{h(x) f(x)}{g(x)} \right]$$

$$z_1, z_2, \dots, z_n$$

$$\frac{h(z_1) f(z_1)}{g(z_1)}, \dots, \frac{h(z_n) f(z_n)}{g(z_n)}$$

$$\hat{y}_{(g)} = \frac{1}{n} \sum_{i=1}^n \frac{h(z_i) f(z_i)}{g(z_i)}$$

$$\hat{g} = \underset{g}{\operatorname{argmin}} \operatorname{var}_g(\hat{y}_{(g)})$$

$$\operatorname{var}_g(\hat{y}_n) = \operatorname{var} \left(\frac{1}{n} \sum_{i=1}^n \frac{h(z_i) f(z_i)}{g(z_i)} \right) = \frac{1}{n} \left\{ E \left[\frac{h^2(z) f^2(z)}{g^2(z)} \right] - \left(E_g \left[\frac{h(z) f(z)}{g(z)} \right] \right)^2 \right\}$$

$$\operatorname{Var} W = E(W^2) - \underline{F}(W)^2$$

$$\text{minimizing } E_g \left[\frac{h^2 f^2}{g^2} \right] = E_g \left[\frac{h^2 f^2}{h^2 h^2 f^2} \right] = \gamma^2$$

$$g^* = \frac{1}{\gamma} h f \Rightarrow \operatorname{Var}_{g^*} = 0 !$$



Problem: We can not use g^* , cause we don't know γ . We have to estimate it.

The main point: The CLOSER the better is, the BEST the solution is. (Acceleration)

Variance reduction via control variable (Slide 28)

$$Z := X + C(Y - m), \quad Y \text{ correlated to r.v. } X$$

$$m = E(Y), \quad C \text{ arbitrary constant}$$

$$\operatorname{var}(Z) = \operatorname{Var}(X + C(Y - m)) = \operatorname{Var}(X) + C^2 \operatorname{Var}(Y) + 2C \operatorname{Cov}(X, Y)$$

$$C^* := - \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(Y)}$$



$$\Rightarrow \text{Var}(t) = \text{Var}(X) - \frac{[\text{Cov}(X, Y)]^2}{\text{Var}(Y)} \leq \text{Var}(X)$$

7. Via Complementary r.v (Slide 29)

Y_1, Y_2 be identically distributed r.v
negatively correlated, mean γ (γ)

The estimator $\frac{Y_1 + Y_2}{2}$ has a smaller variance than that with the independence assumption.

$$\text{Var}\left(\frac{Y_1 + Y_2}{2}\right) = \frac{1}{4} [\text{Var}(Y_1) + \text{Var}(Y_2) + 2 \text{Cov}(Y_1, Y_2)]$$

8. Via Conditioning (Slide 30)

Regression:

$$E[X|Y]$$

$$E[X|Y=y] = h(y)$$

$$E[X|Y=y] = \int x f_{x|y}(x|y) dx = h(y)$$

$$x \mapsto F_{x|y}(x|y) = P(X \leq x | Y=y)$$

$$x \mapsto f_{x|y}(x) = \int f_{xy}(x, y) dy$$

$$\rightarrow E[E(X|Y)] = E[h(Y)] = E(X)$$

Total expectation formula

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)]$$

Total variance formula

$$\text{we get that } \text{Var}(X) \geq \text{Var}[E(X|Y)]$$

9. Implementation (Slide 32)

h { - math. function
- algorithm
- procedure

$\Rightarrow$

STEPS:

- 1) Method to sample the r.v
- 2) Calculate $h(X_i), i=1, \dots, n$
- 3) Define the precision (ϵ, θ)
- 4) Define n in order to achieve precision
- 5) Determine g if necessary
- 6) Calculate:

$$\frac{1}{n} \sum_{i=1}^n h(Y_i) \frac{f(Y_i)}{g(Y_i)}$$

10. Bootstrap

WHY? Increase the data.

Before bootstrap:
Conditional Expectation

ex 1:

N times r.v $N^* = \{1, 2, \dots\}$
 $S = X_1 + X_2 + \dots + X_N$ r.v

$$\begin{aligned} 1. \quad E(S) &= E\left[\sum_{i=1}^N X_i\right] \\ &= E\left[E\left(\sum_{i=1}^N X_i \mid N\right)\right] \\ &= E\left[\sum_{i=1}^N E(X_i \mid N=n)\right] = E\left[\sum_{i=1}^N E(X_i)\right] = n E(X_i) \end{aligned}$$

$$2. \quad E(S) = E[N \cdot E(X_i)] = E(X_i) E[N]$$

ex 2:

$y_1, y_2, \dots, y_n$ sensors
 X state of the motor
 $Z = \gamma(y_1, \dots, y_n)$ close to $\underline{X}$
 $\arg\min_{\gamma} E[(X - \gamma(y))^2] = \gamma^*(y) = E[X|Y]$

($\gamma \rightarrow$ function we are looking for)

Proof:

Prove that for any γ $E[(X - \gamma(y))^2] \geq E[(X - E(X|Y))^2] \geq 0$

$$E[(X - \gamma(y))^2] - E[(X - E(X|Y))^2] \geq 0$$

$$E[\cancel{X^2} - 2X\gamma(y) + \gamma^2(y) - \cancel{X^2} + 2XE(X|Y) - E^2(X|Y)] \geq 0$$

$$E[-2X\gamma(y) + \gamma^2(y) + 2XE(X|Y) - E^2(X|Y)]$$

$$E[Xh(X)|Y] = h(Y)E(X|Y)$$

$$-2\gamma(y)E[X|Y] + \gamma^2(y) + 2E[X|Y]E(X|Y) - \cancel{E^2(X|Y)}^2 \geq 0$$

$$= \gamma^2(y) - 2\gamma(y)E[X|Y] + E^2(X|Y)$$

$$[\gamma(y) - E(X|Y)]^2 \geq 0$$